

SATURNE: food at the heart of a sustainable and human network

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ABSTRACT

Humans are more than ever disconnected from their food. The gap between consumers, food and producers is growing. As a result, customers rarely ever express their gratitude to producers. The food industry appears to have lost some of its fundamental values. This paper aims at providing a better understanding of the distances that separates consumers, their food, and producers. Most importantly, it will allow for moving beyond harmful practices that are currently pervasive in the food industry. We explore the possibility of recreating meaningful and deep connections between the different actors and food.

CCS CONCEPTS

• **Human-centered computing** → Collaborative and social computing; Collaborative and social computing design and evaluation methods; Ethnographic studies; Interaction design.

KEYWORDS

food, communities, environment, ethnographic design, interaction design, Quebec

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1 INTRODUCTION

From the first public markets to the most recent online purchasing practices, food in Quebec has undergone many changes. Without a doubt, the most recent upheaval is the pandemic and the post-pandemic context, in which we must now revisit many gains that have been undermined. The food industry has been strongly affected and paradigm shifts seem more and more imperative. Political, environmental, social, and economic issues are intimately linked to the food system, making the topic complex and, above all, more important than ever.

From the point of view of positive psychology, gratitude is still to this day an emotion which is not fully understood by science. Positive psychology is defined as the study of the conditions and processes that contribute to the flourishing and optimal functioning of groups, communities, and institutions [1]. More specifically, gratitude is defined by the Oxford dictionary as "the quality or condition of being thankful; the appreciation of an inclination to return kindness [2]. Gratitude is, first and foremost, a social emotion [3].

In the context of food, gratitude is an expression of appreciation for nature, which provides us with products, and for those who produce or grow them. It is about being fully aware of what is offered to us. Thus, responsible, and sustainable practices are derived from such an approach to food. However, to this day, there is a marked disconnect between food and consumers, and between consumers and producers, in the food buying experience. This experience is increasingly impersonal. Consumers no longer have any idea where their food come from or who helped bring it into their homes. Never have consumers been so disconnected from their food [4]. We once knew all the work that went into food production. This is no longer the case. The COVID-19 pandemic has greatly shaken the food industry. It has left individuals with no choice but to realize the fragility of our food system and its supply. Shutdowns, lockdowns, and curfews. . . The world was turned upside down. With no other options, businesses, followed by their consumers, turned to digital platforms to ensure their survival, once again removing the human and sensitive touch of food [5]. The food's history was even more disjointed, the producer erased and the consumer more alone than ever. However, we believe there is more and more palpable interest of consumers towards local purchase, responsible consumption, sustainable development, etc. [4]. Thus, there is a gap between current practices and this interest in responsible food. A gap that appears to be widening rapidly.

Following these initial reflections, our team's main research objective was to understand the factors influencing gratitude during the purchase of unprepared food products by exploring and observing expressions of gratitude during the different phases of the food purchasing experience to promote the feeling of gratitude and its expression among all the individuals involved. This objective is part of the overall mandate we have been given, which consists in proposing a solution that favors the expression of gratitude through design. The whole project needs to be based on human-centered design approaches and to allow the development of new vectors of gratitude.

Consequently, the team has defined six (6) research objectives: 1- Understand the ecosystem of the food buying experience and the different actors it includes. 2- Understand the different phases

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of online shopping. 3- See if there is a link between the level of abstraction of the service and the perception and feeling of gratitude of the user. 4- Understand the factors of understanding and defining a service and its tangibility. 5- Determine if there is a temporality associated with the feeling of gratitude. 6- Determine whether there are elements in the service ecosystem that are recipients of gratitude by the user.

2 METHODOLOGY

Within the framework of the project, our team oriented its design process according to two approaches. First, the research stage was conducted according to the ethnographic approach applied to design but began with a literature search. The model of the ethnographic approach applied to the design (Design ethnography timeline) of Nova guided our process [6]. We also tried the experiment ourselves in different types of food businesses. The second part of our approach is based on the human-centered design approach. The approach is therefore innovation-oriented and iterative. It corresponds to the Understand, Design, Evaluate model.

2.1 Participatory and non-participatory observation

The field research began by observing user behaviors and connections made with food. This was done both online and in physical locations. To do this, our team went to different types of shops, homes, and other types of locations. “Participant observation was the fundamental approach to gaining insights into consumer behavior. It was used to develop our understanding of how people shop and use products in their daily lives. Researchers might roam the aisles of a grocery store, identifying characteristic browsing behaviors” [7]. Then, to further explore the experience, we observed the participants and asked them to narrate what they experienced, their thoughts and their impressions. At this point, we focused on analyzing the moment of reception of the ordered goods on a food shopping website. This provided privileged access to the individual’s actions, thoughts, difficulties, and emotions.

2.2 Surveys and interviews

In the second phase of the research, we conducted a non-probabilistic sampling including 30 participants who were to complete a questionnaire and undergo an interview process. The participants were divided into two groups. The first group was asked to answer the questionnaire, while the second group took part in an informal interview about their eating habits. The activity lasted between 15 and 30 minutes. The main objective of the questionnaire was to understand the ways in which users value certain types of food on their lifestyle. It was constructed based on IDEO research methods, namely, the personal inventory and the five whys [8]. The sampling for the interview was semi-directed and included participants living outside of Canada. The primary goal of the survey was to gather information about different cultures to further our understanding of the situation.

3 RESULTS (UNDERSTAND)

Based on the analysis of our research data, our main finding is that there are recurring factors that promote gratitude in the food buying

experience. These factors represent features of the experience that contribute to the consumer feeling and expressing gratitude. The factors identified had to be recurrent among different types of users and their varying purchasing practices. They include: 1- the feeling of freedom of choice 2- the ability to attribute tangible characteristics to the food (ex. I can see it, I can touch it, I know where it comes from. . .) 3- the feeling of belonging to a community 4- an authentic human contact that sets aside sales practices 5- respect for the environment (through the entire service ecosystem) 6- knowledge about the product 7- the ease and speed of obtaining the product; and 8- accessibility for all. From this analysis, we were able to correlate the criteria with what we had observed and heard in the field. The main finding is that expressions of gratitude are virtually non-existent in many food services due to major flaws in the consumers’ experience.

3.1 Design opportunities

This reality, once again very present between theory, needs and practice, allowed us to target the design opportunities that would serve as an opening to our intervention. Four were selected.

The first design opportunity lies in the fact that constraints are omnipresent in the shopping experience, which hinder the autonomy and freedom of the user. Indeed, the user is framed against his will in almost all his shopping experience by 1- the inevitable moments of waiting 2- a pressure to get everything done quickly and 3- a narrow and non-transparent selection process of the information he is presented about the products. This situation translates into a lack of control that the user has over the course of their food shopping experience. The second opportunity stems from the food industries’ lack of involvement in sustainable and ecological practices. Important information about the environmental impact of products throughout their life cycle is not sufficient to meet the informational needs of the user, who lack the means to make an informed choice regarding environmental protection. In this regard, the trend is currently increasing in terms of knowledge of the origin of food. Traceability is becoming an issue as the population knows less and less about the origin of what it consumes [9]. Third, food is currently treated as a marketing product when the consumer’s needs are quite different. There is a gap between the capitalist intentions of companies and the vision of consumers. Instead, consumers perceive food as meeting an essential need, as a source of vitality, and as a highly meaningful product. The ubiquitous sales practices that surround food thus undermine the connection between the consumer and his food. Finally, the last design opportunity is rooted in human contact. There is a conflict between the brief and impersonal human contact that takes place in today’s retail environment (physical or digital) and the social and sensitive needs of users. In users’ shopping experiences, human contact no longer exists for what it should be, but rather is a marketing tool for brands.

4 SOLUTION (DESIGN)

The proposed solution, *Saturne*, is a brand-new socio-technical food system that allows the transmission and standardization of information in addition to the tracking of food products. *Saturne* also creates a community of exchange and mutual aid in the food

industry. The solution also serves as a vector for awareness and citizen mobilization by making responsible and sustainable practices accessible and simple.

Our solution includes a physical and a digital component. The physical component is the heart of the solution. We first concentrated our efforts on this component because we believed that a low-tech solution was consistent with the environmental values we wish to convey. We care that our solution is supported by practice that have little impact on the environment. Moreover, current trends push individuals to the inevitable use of technology, without them knowing exactly why. "We've built a technology-focused society that is remarkable on means, but hazy about ends. It's no longer clear to which question all this stuff-tech-is an answer, or what value it adds to our lives. Too many people I meet assume that being innovative means adding technology to it" [10]. However, our research has highlighted the importance of the digital shift which is accentuated by the post-pandemic context. Thus, we opted for a digital component requiring few technological resources. The two components are closely linked and complementary but can nonetheless be used independently.

4.1 Redesign and standardization of information

The first part of the Saturne project consists in reinventing and modernizing food labels. It uses standardized information tool that is already well-known and widespread to review all the information it presents.

Based on the previously identified gratitude criteria and the targeted opportunities, we were able to determine the relevant information to be presented on the label. The main goal is to offer information that corresponds to the current needs of consumers to allow them to carefully choose their products.

To achieve this, we have created three indexes that serve to group information according to its content. The order in which specific items of information appear on the label is random and does not imply a hierarchy in terms of importance. We consider them equally important and prioritize free interpretation by the user. These indexes are also based on standardized portions (quantity of the product) to facilitate the comparison between different products. Thus, for a product of the same nature (e.g., liquid), the portions analyzed will be equivalent. The Health Index is a direct reflection of the Nutrition Facts table. It is an important element for consumers. However, to make this information easier to understand, we have standardized and simplified it. We therefore imagine a codified system, easy to learn and understand by all. The second index is the Environmental Index. It directly addresses the second design opportunity by providing information about the environmental impact of the product in a transparent way. At this point, our research has identified the amount of land required, the amount of carbon dioxide emitted and the water footprint as the most relevant information to display. The third index is the Human Index. It serves to create a strong link between the producer and the consumer. Information such as the name of the producer or the farm, the place of origin, the number of people who contributed, etc.

We also add a section to insert a personalized message from the producer so that he can present his product to the consumer.

The production and expiration dates are highlighted more than on current products, where they are often scattered. The information presented makes the work behind the product and its positive impacts much more tangible to foster a feeling of gratitude. This feeling can be directed towards the producer now made visible. The entire label is presented in black and white not only to display a neutral appearance, but also to avoid using additional material such as ink that would be unnecessary to the message.

The final element added to the revamped label is the QR code that allows a direct and accessible link to the digital component of the solution. An issue arises with respect to unpackaged products. As a result of our guiding design values, it is important that we do not encourage over-packaging practices. The QR code could be laser-etched into fruits and vegetables to limit the use and production of plastic stickers, which often end up in the compost. The label could then be placed directly on the displays or shelves. Note that this QR code is also found on the bill obtained at the cash register to integrate a more advanced and rapid analysis and visualization of its food consumption in addition to offering recipes based on the Canada Food Guide.

4.2 Formation

To meet the informational and social needs of the consumer, our solution includes a considerable effort on the training of food specialists. Specifically, on site, we aim to bring the human face back to the contacts made by making the interactions meaningful, deep, and useful. In this way, we add new faces to gratitude through human contact based on sharing deep knowledge, mutual support, and authenticity. Here, we restore value to the human connection that is so necessary, especially in the context of the isolation created by the pandemic.

4.3 Application development

By scanning the QR code on the products and the invoice with your smartphone, you can easily access the Saturne digital platform. The platform is a hybrid application. It can be accessed via the web browser and can also be

installed on a mobile device. First, there are detailed food product sheets. These cards are made up of the same type of information as the label, but the digital nature allows us to go deeper.

Several features that promote the feeling of gratitude and its expression between individuals are also integrated. For example, it is possible to express our gratitude to the producer of the product with quick reactions. To encourage the creation of a real community, it is important to allow the creation of interpersonal links. The user can therefore send a message to the producer to express his gratitude by thanking him, congratulating him, etc. We also want to encourage exchange and a sense of community through sharing features. It is therefore possible to share a product, a place or a favorite producer with our friends and family. Producers receive, in real time, all the shares, reactions and messages, to create a brief temporal space in which gratitude is experienced simultaneously.

The solution unites the consumer and the food throughout the life cycle that follows the purchase. Thus, from one or more selected products, it is possible to access local recipes that highlight these food items. The application will allow the user to exchange

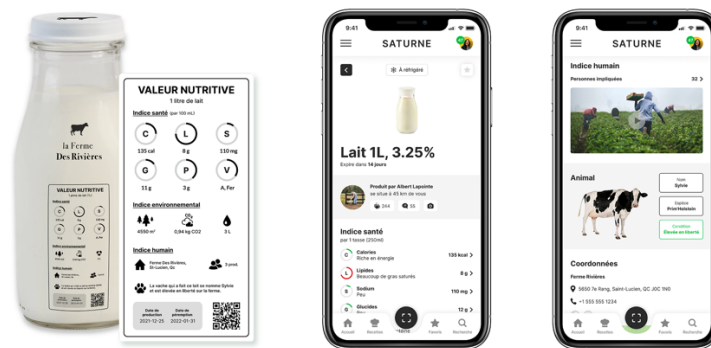


Figure 1: – The physical and digital components of Saturne

with a food specialist to obtain information, discover new products, explore nearby production sites, etc. Through geolocation, the food can be easily traced to establish the distance traveled by the product to reach the consumer. The producer also has access to this information to be aware of what happens to his products.

To ensure the proper functioning of the platform, producers have a role to play in feeding the application with information. Some information will be mandatory, but others will be optional. Thus, forms will be available for them to enter information to share about their product. The whole process will be made easy and short by clear interactive forms in which the producer will be guided if he wishes.

We consider several systems to be inseparable from the success of our solution. Thus, the political system becomes an important part since it governs the norms and standards related to food. Its importance lies in the essential needs for the deployment of the solution. As for the economic system, it is concerned by the impacts of our solution. Thus, we aim, gradually, to make a transfer towards a more sustainable and circular economy. The whole is explained by the responsible and significant purchase, a much healthier production, and a return to the community. Obviously, Quebec's social systems are at the heart of our approach since we use them as a resource, but also as a goal. We strengthen them by promoting practices of mutual aid, exchange, and support.

5 DISCUSSION (EVALUATE)

5.1 Related works

Our solution is part of a context of change. We are currently witnessing changes in the food industry. Various research studies and initiatives have helped contextualize our solution and give us direction. For example, digital platforms such as Maturin and Terroirs Québec, have as their main objective to create links between consumers and producers [11, 12].

Other trends related to our approach are those of conscious eating and responsible eating. This type of practice, which is being explored more and more, consists of taking the time to be fully aware of our food, but also of what surrounds it. Specifically, it is eating slowly, with gratitude and attention, avoiding waste, favoring fresh, unprocessed, and organic foods, etc. [13]

The International Reference Centre for the Life Cycle of Products, Processes and Services [14] has a component entirely dedicated to the food industry. Their approach focusing on analysis, innovation and technologies is enriching for our project. The “Pôle d'excellence Agroalimentaire” (PEA) aims to develop skills in the analysis of complex and innovative agricultural systems. Through a holistic approach, the PEA addresses the question of sustainability by focusing on issues that closely match our work, such as the eco-efficiency of food systems, the use of tools for traceability and the development of innovative methodologies [14].

Although outside the scope of our work, the American project The Farm Project is relevant to our project. They denounce the lack of knowledge about our food products as well as the excessive disconnection between humans and their food. The current broken system is a big part of the problem. The Farm Project aims to reconnect people with their food: “We aim to empower people to ask the right questions - because when we ask the right questions, we can make informed choices and vote with our dollars for a better food system. [14]”

5.2 Participatory evaluation approach

In addition to building on such initiatives that are larger than ours, our solution also draws credibility from the evaluation that has been done. Thus, in the final stages of our process, we conducted a participatory evaluation. The process was carried out with five (5) participants and lasted about twenty minutes. The evaluation was conducted online. First, we presented a brief description of the project and our solution. Then, the participants were asked to perform a series of tasks which included reading and understanding

the label as well as using the digital platform. The participants had to perform the tasks and verbally describe what they were thinking and doing. We observed them simultaneously. The evaluation allowed us to validate both the system as a whole and its foundations as well as more specific factors such as the application's functionalities and the codes/indices developed for the product sheets. Our participatory evaluation approach allowed us to correct some shortcomings. These were mainly related to the difficult understanding of the iconographic system used to represent the different information on the products. For example, the icon representing the water footprint did not make it possible to identify the nature of the information. We then modified our icon system to ensure a better understanding. Fortunately, our solution was very well-received by the participants, who unanimously supported its objectives. The issues which motivated our intervention resonated strongly with the participants, who identified with the project.

5.3 Project status

Our team is aware that a solution of this magnitude requires the involvement of many additional resources. Considering our field of expertise which is design, associations with scientists and government institutions would be inherent to the viability of the project. Thus, in addition to the support obtained by our participants, the faculty of our university and the valuable contributors of the research on which our intervention is based, specialized external resources would be necessary. To carry out such a project, it would be necessary to consult food specialists such as nutritionists to ensure the accuracy of the Health Index. In addition, it would be essential to conduct further scientific research to select the most important environmental factors more accurately in the field. This could be possible through partnerships with Quebec research centers. Finally, the involvement of the Canadian government is essential to standardize food information and the legislation that surrounds it.

6 CONCLUSION

In conclusion, Saturne is a socio-technical system that addresses the problem of consumer disconnection from the producer and the food giving control of the experience back to the actors involved. The solution renews the food buying experience by creating meaningful connections between producers, consumers, and food. Gratitude is lived and felt in parallel. Each step of the solution is a vector of gratitude. The low-tech solution allows us to concretely transmit the ecological and sustainable values of our vision while meeting the informational and social needs of users. Through this tool, we promote a real commitment by consumers to sustainable, ethical, and conscious food practices. We do this by providing transparent information and creating an authentic community. We are confident that this type of innovation can contribute to creating positive change in our society and its environment. Saturne is a comprehensive solution that, through the creation of the Health, Environmental, and Human Indexes, considers the most important issues in the food industry. With the exponential advances in technology, the possibilities before us are numerous. With the protection of the environment in mind, simple and ecological technologies could very likely be an avenue to consider.

For example, data collection systems at the source allow for better management of our entire food system. Currently, the connected farm model would, in theory, feed a global population while using fewer resources. Specifically, the data collected is used to simulate agricultural processes and analyze them in depth to adapt them to the needs of the population [9].

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